

Centralized MAC protocols

MAC Protocols

- Previous lecture – distributed contention based MAC protocols, users decide who should transmit when in a decentralized manner
- Today's lecture – a central entity allocates resources to users sharing a medium
 - **TDMA** – Time Division Multiple Access
 - **CDMA** – Code Division Multiple Access
 - Other concepts – **SDMA**, **FDMA** / **OFDMA**
- Mainly used in cellular networks, as voice requires high QoS.

TDMA

- Assign different time slots to different users



- Fixed TDMA – Each user gets a fixed time slot irrespective of whether he has data to send or not
 - Wastes slots when users have bursty data
- Dynamic TDMA – the decision of which user sends when is decided on a per-slot basis
 - Users signal their intention to send data
 - Slots allocated to users who have data to send
 - Scheduling algorithm decides which user is scheduled to transmit in which slot

TDMA



- Synchronization
- Perfect for constant data rate, very inefficient for asymmetric or bursty data, good for voice, too inflexible for data

Scheduling algorithms

- Dynamic TDMA relies on scheduling algorithms. Tradeoff between efficiency and fairness.
- Common scheduling algorithms used in cellular networks
 - **Round robin** – schedule all users in a certain order. Guarantees fairness.
 - **Max rate** – schedule the user that has best channel conditions, i.e., can send at highest rate. This guarantees that the network gets high throughput. But may starve some users at cell edge.
 - **Proportionally fair** – schedule users according to a priority computed as $p = \text{current_rate} / \text{average_rate}$. The current rate is computed based on current channel conditions. So biased towards users with good channel and high rate. Also avoids starving of some users, because if average_rate becomes low enough, the user priority will increase and he will get scheduled.
- Proportionally fair scheduler (or its variants) is the most common design used in today's networks.

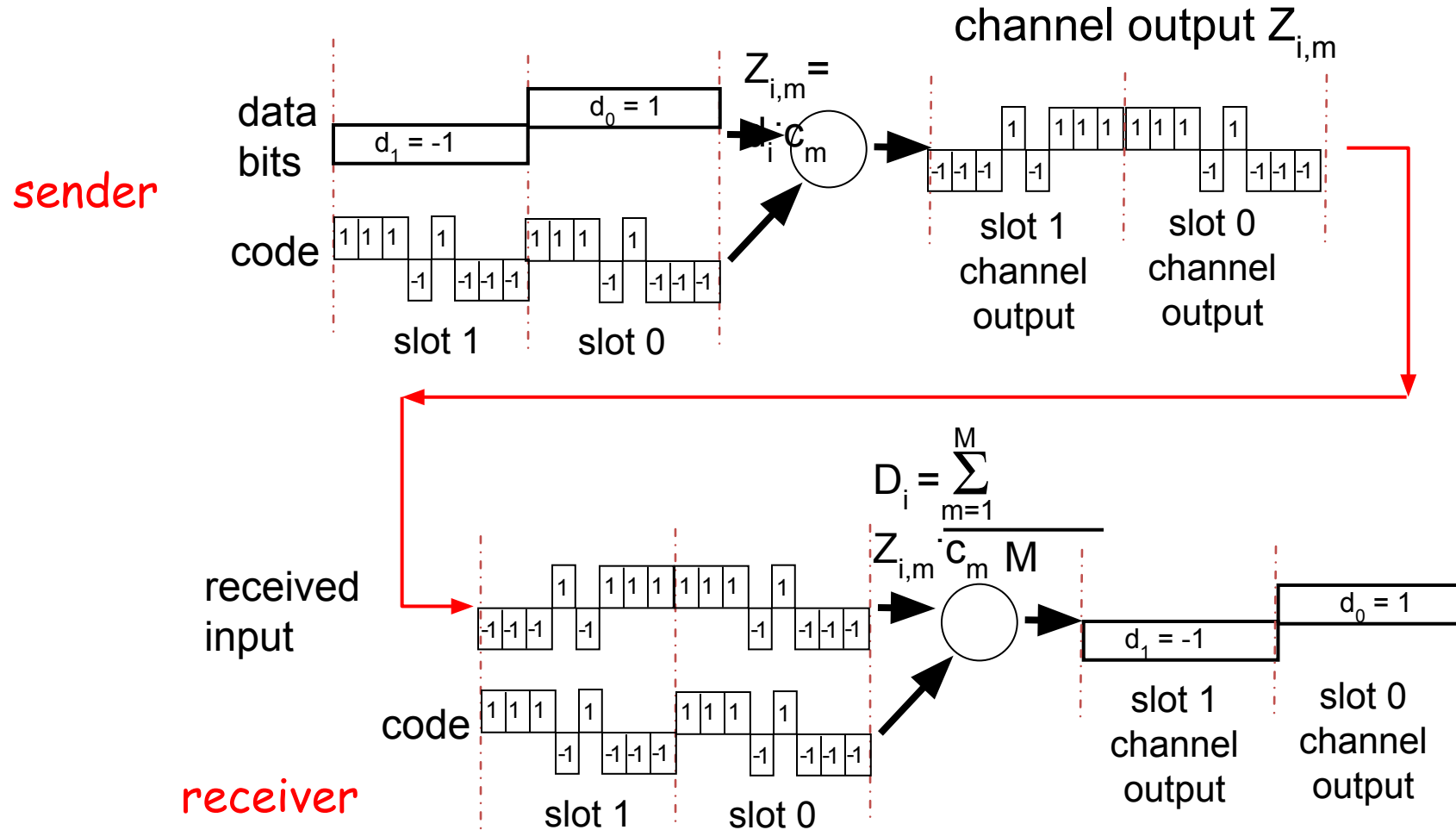
CDMA

- Basic idea: transmit each user's data using a **unique code**.
- Take each bit, exor with a longer bit sequence called code, and transmit the resulting new bit stream.
 - For example, suppose a user's code is 010011. Then, for bit 1 the user sends the code "010011". For bit 0, the user sends the complement "101100".
- At the receiver, correlate with the code to recover data.
 - If correlation with 010011 is high, then it is 1. If correlation with complement is high, then it is 0.
- Different users are assigned different **"orthogonal"** codes (inner product is 0), that is, codes which have low correlation with each other.
- Even if the signals of multiple users are combined, the receiver can extract its own transmission by **correlating with its own code**

Code Division Multiple Access (CDMA)

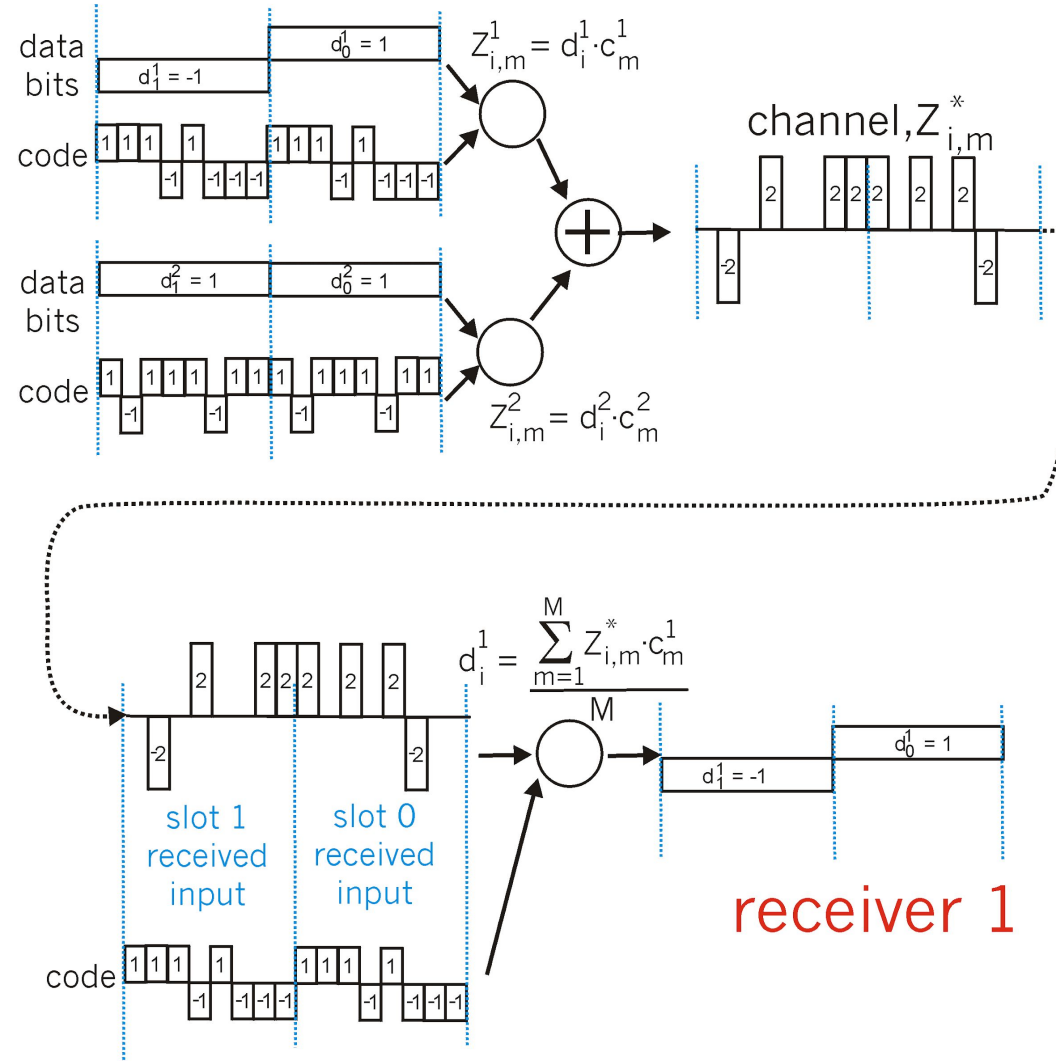
- Used in several wireless broadcast channels (cellular, satellite, etc) standards
- *Encoded signal* = (original data) X (chipping sequence)
- *Decoding*: inner-product of encoded signal and chipping sequence
- Allows multiple users to “coexist” and transmit simultaneously with minimal interference (if codes are “orthogonal”)

CDMA Encode/Decode



CDMA: two-sender interference

senders



CDMA (2)

- Example, user A uses code 010011 and user B uses code 110101
- Suppose A wants to send bit 1 and B wants to send bit 0. Let's assume we send -1 for code bit 0.
- A sends $(-1, 1, -1, -1, 1, 1)$ and B sends $(-1, -1, 1, -1, 1, -1)$
- In a simplistic model where both signals combine, we get $(-2, 0, 0, -2, 2, 0)$
- Correlate received signal with A's code gives $+6 \rightarrow$ bit 1
- Correlate with B's code word gives $-6 \rightarrow$ bit 0
- If B's transmit power is much higher than A's, that is, B's bit sequence is scaled up, then harder to decode A's bit.
- Power control is very important in CDMA, as other transmissions appear as noise and reduce SNR

Frequency Domain View of CDMA

- Multiplying a bit with a code is equivalent to spreading the spectrum in the frequency domain (recall: faster pulses -> wider bandwidth)
- That is, each user uses a larger bandwidth than the original signal
- However, CDMA is not inefficient because many users are multiplexed over the same wider band

Other ways of multiplexing

- **Frequency Division Multiple Access (FDMA)** – assign multiple narrow channels to different users.
 - Channels can be assigned same frequency at all times
 - Change frequencies according to a certain pattern, frequency hopping
 - Tx and Rx agree on a hopping pattern
- **Orthogonal Frequency Division Multiple Access (OFDMA)** – Similar to OFDM, but different sub carriers can be allocated to different transmitters.

Other ways of multiplexing

- **Space Division Multiple Access (SDMA)** – the idea behind having “cells” in cellular networks. Frequencies used in one cell can be reused in another cell that is some distance away.
 - Allocating a separated space to users
 - Typical app: optimal base station to a mobile phone user, taking into account which frequencies (FDM), time slots(TDM), code(CDM) are still available
 - Typically, SDM is not used in isolation, always in combination with one or more other schemes

Challenges in centralized MACs

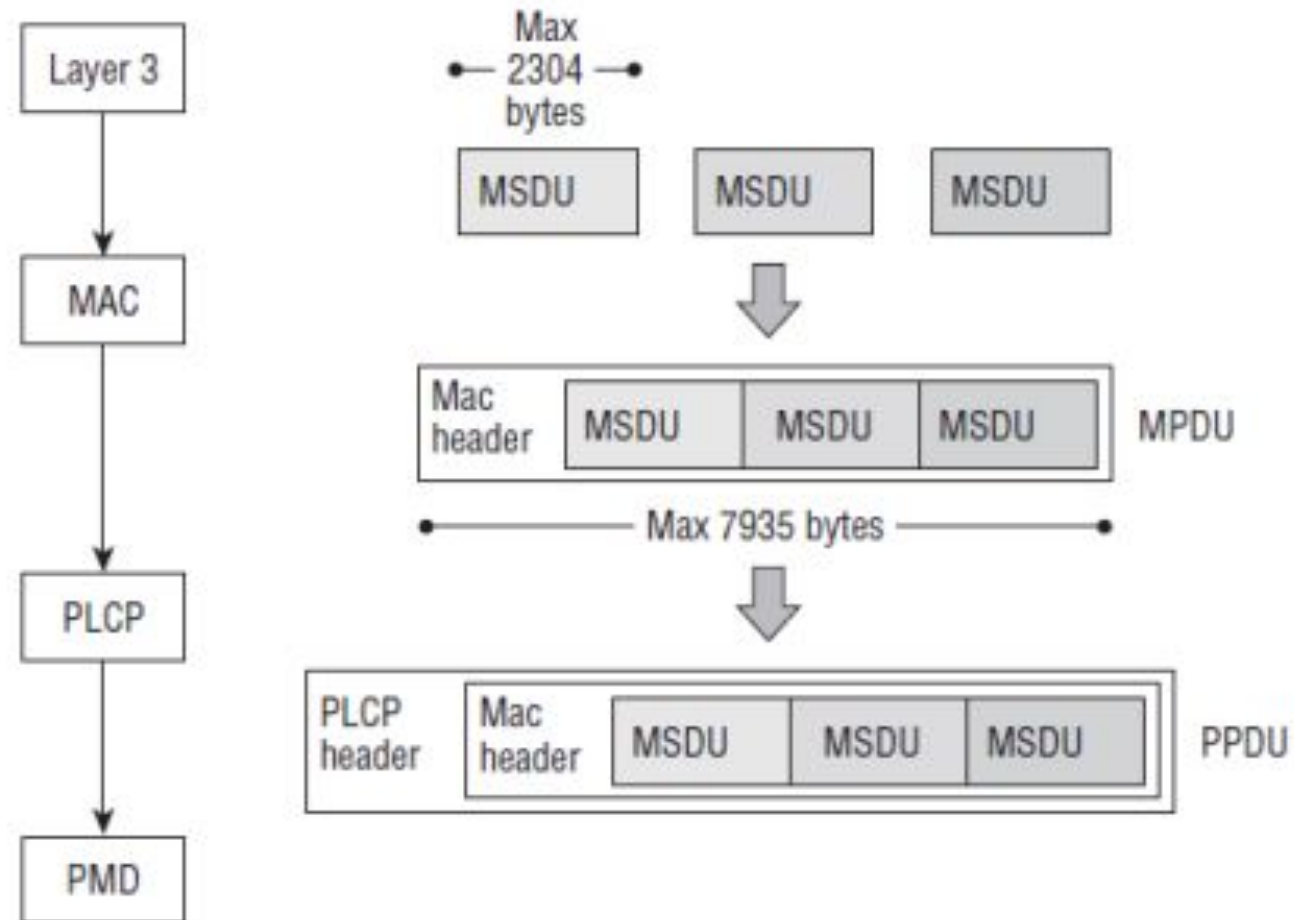
- TDMA requires tight time synchronization
- CDMA requires fine-grained power control (and possibly time sync)
- FDMA requires very precise channel filters to restrict users to specific frequencies

Multiple Access in Cellular Networks

- All **cellular networks** use **SDMA** to partition frequencies to cells
- **2G** networks mainly used plain **TDMA** (in GSM networks) or **CDMA** within a cell
- **3G** networks use a combination of **TDMA** and **CDMA** in a cell
 - Voice mainly uses CDMA
 - Special high speed data channels exist in some 3G technologies. These use a combination of TDMA and CDMA. In every slot, a single user or multiple users can be scheduled. If multiple users, they are multiplexed using different codes.
- **4G / LTE** uses **TDMA + OFDMA** on the downlink. That is, in each slot, a single user can be scheduled, or multiple users can be scheduled over multiple subcarriers in OFDM.

Frame Aggregation

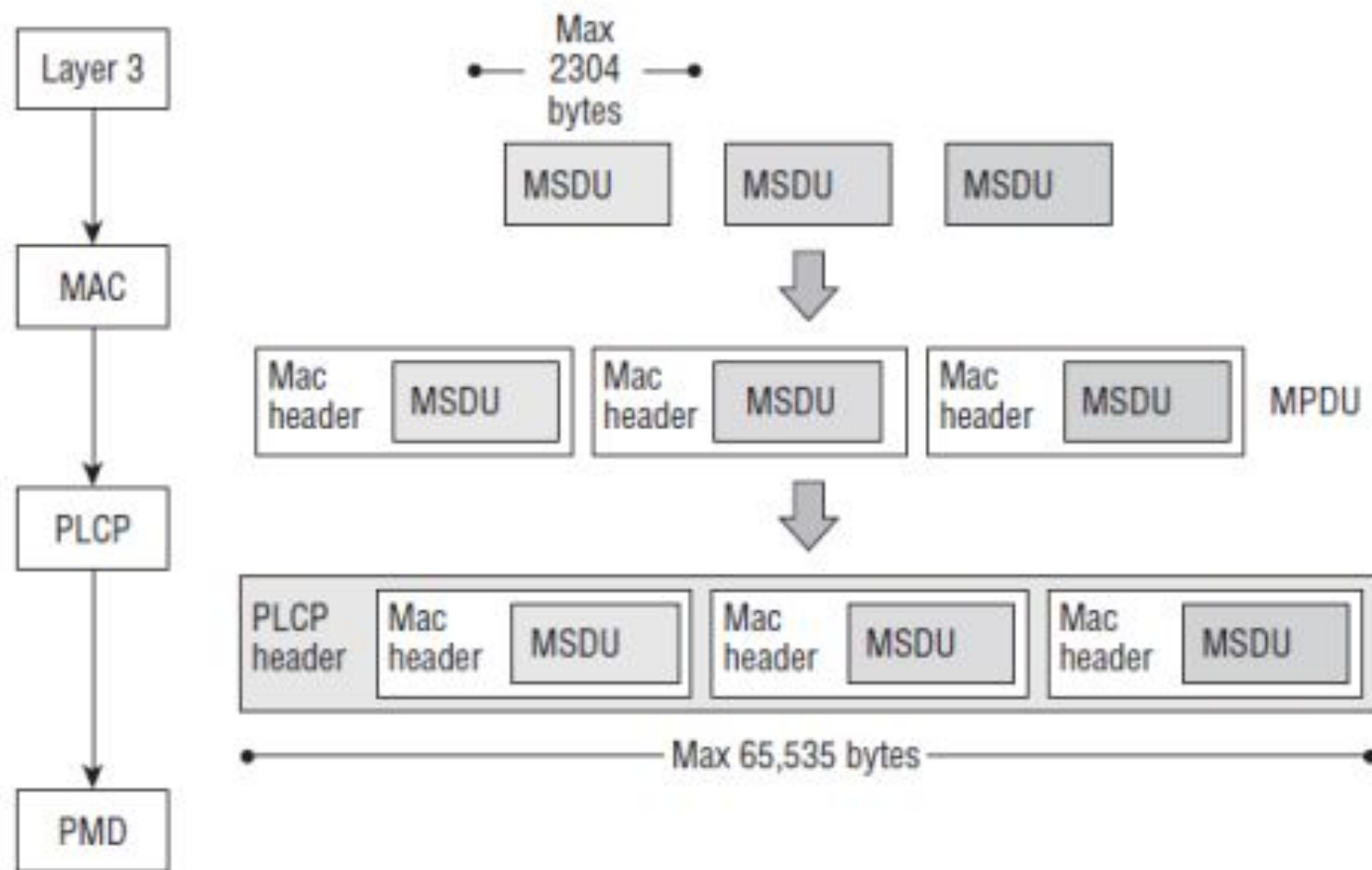
FIGURE 6.8 A-MSDU frame aggregation



A-MSDU aggregation

- With A-MSDU, the maximum frame body size is determined by the maximum A-MSDU size of 3,839 or 7,935 octets, depending upon the STA's capability, plus any overhead from encryption.
- The fixed MAC layer overhead is reduced, and overhead caused by the random backoff timer during medium contention is also minimized.
- Multiple MSDUs can be aggregated into a single frame transmission.
- The aggregated MSDUs will have a single wireless receiver when wrapped together in a single frame.
- **Individual MSDUs must all be of the same 802.11e QoS access category.**
- Voice MSDUs cannot be mixed with Best Effort or Video MSDUs inside the same aggregated frame.

FIGURE 6.9 A-MPDU frame aggregation



A-MPDU aggregation

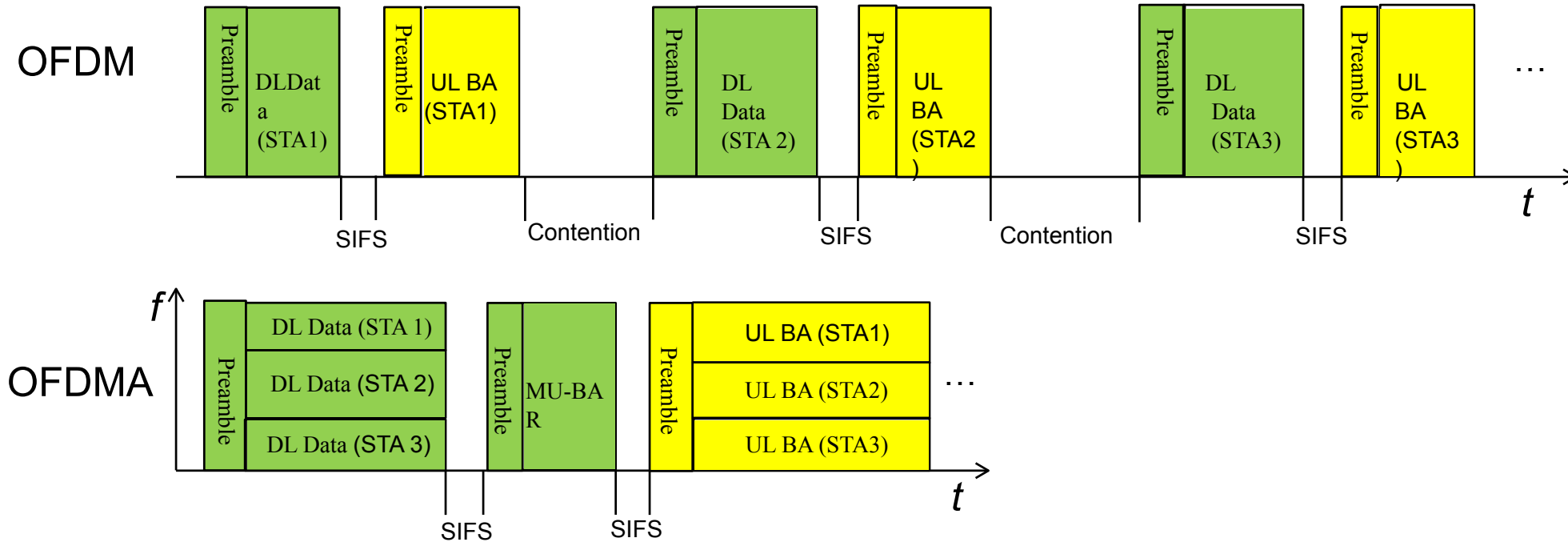
- The individual MPDUs within an A-MPDU must all have the same receiver address.
- Like MSDU aggregation, individual MPDUs must all be of the same 802.11e QoS access category.
- Voice MPDUs cannot be mixed with Best Effort or Video MPDUs inside the same aggregated frame.
- MPDU aggregation has more overhead than MSDU aggregation because each MPDU has an individual MAC header and trailer.
- CRC errors can be detected in the individual MPDU frames and therefore an entire A-MPDU does not need to be retransmitted, only the individual MPDU that is corrupted. Therefore, A-MPDU is less susceptible to noise than A-MSDU.
- **The majority of WLAN vendors use A-MPDU.**
- **All 802.11ac frames are transmitted using the Aggregate MAC Protocol Data Unit (A-MPDU) frame format, even if only a single frame is being transmitted.**
- Max A-MPDU length: 65535 bytes 802.11n 1MB for 802.11ac
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Block Acknowledgement

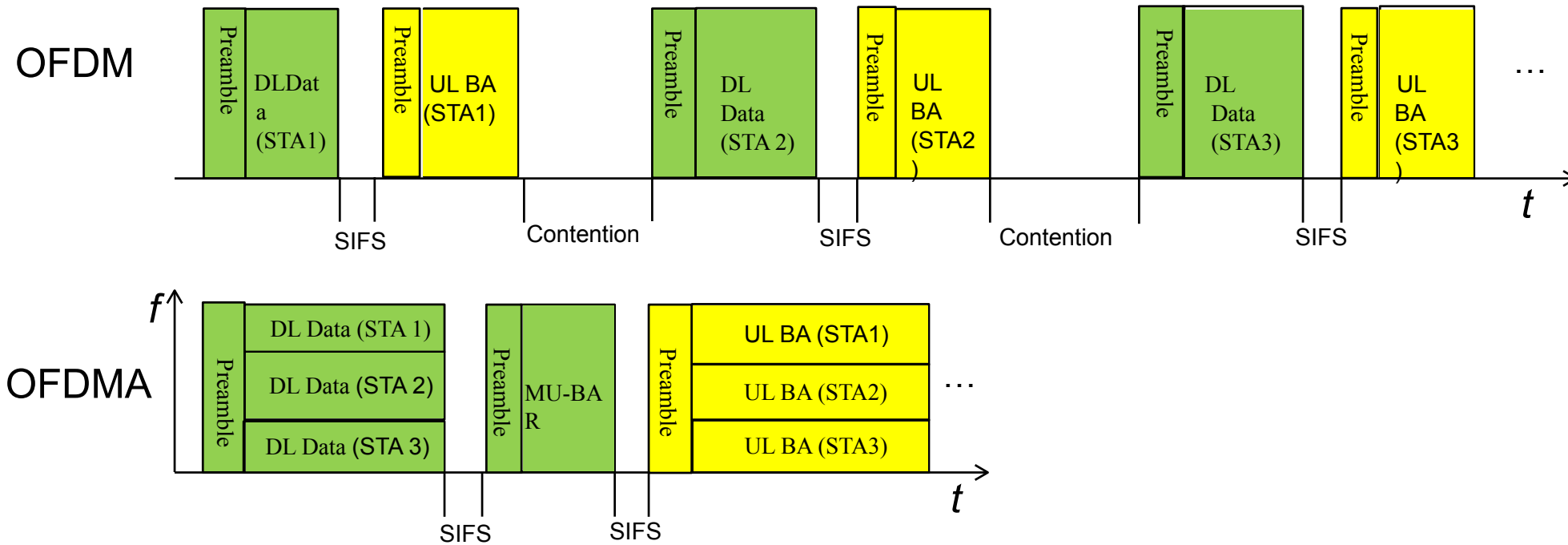
Instead of transmitting an individual **ACK** for every **MPDU** (i.e., **frame**), multiple MPDUs can be acknowledged together using a single BA frame. Block Ack (BA) contains bitmap size of 64×16 bits. These 16 bits accounts the fragment number of the MPDUs to be acknowledged. Each bit of this bitmap represent the status (success/failure) of a MPDU.

OFDMA

Orthogonal Frequency Division Multiple Access

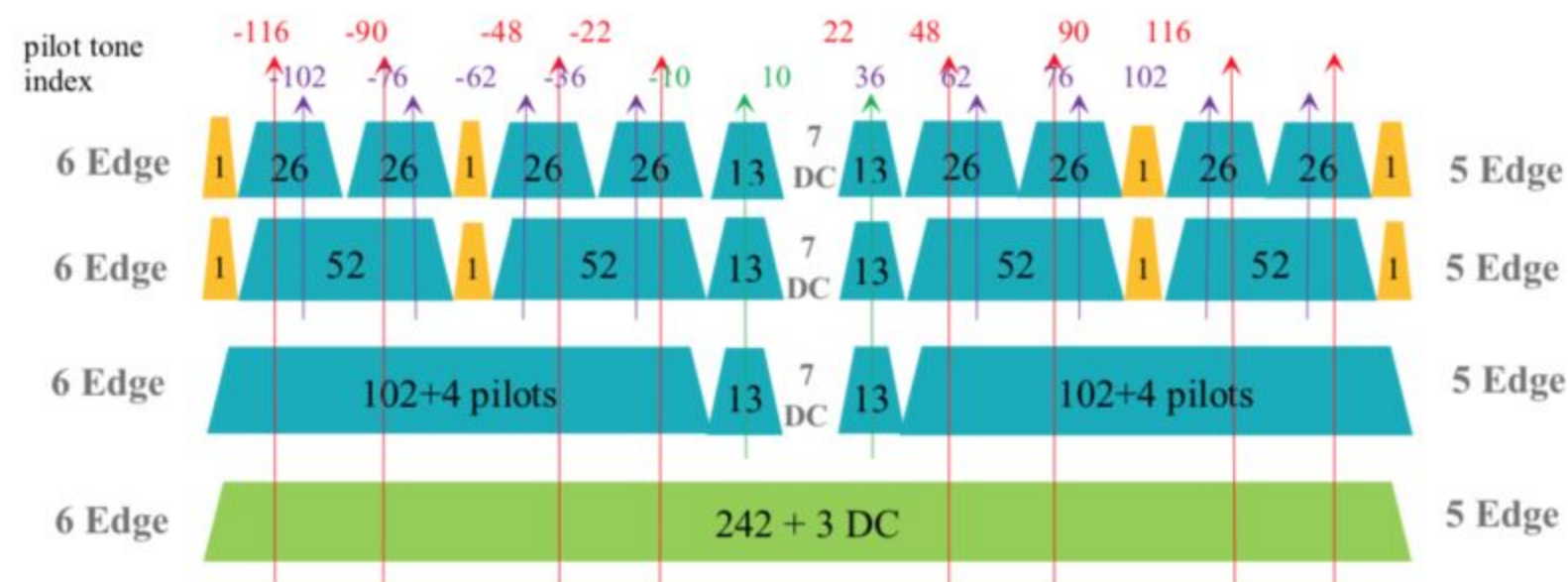


Orthogonal Frequency Division Multiple Access



- Issue: MAC efficiency drops, increased STA density and short packets (more contention, collision, IFS, preambles)
- **Aggregation** in 11n combines short packets in **TIME** from a **single** user, **OFDMA** combines **different** users together in **FREQUENCY (RU)**
- OFDMA does **NOT** increase the maximum PHY rate
- Downlink OFDMA: AP groups users to maximize downlink transmission efficiency
- Uplink OFDMA: Users are grouped together and transmit in sync to AP to maximize uplink transmission efficiency

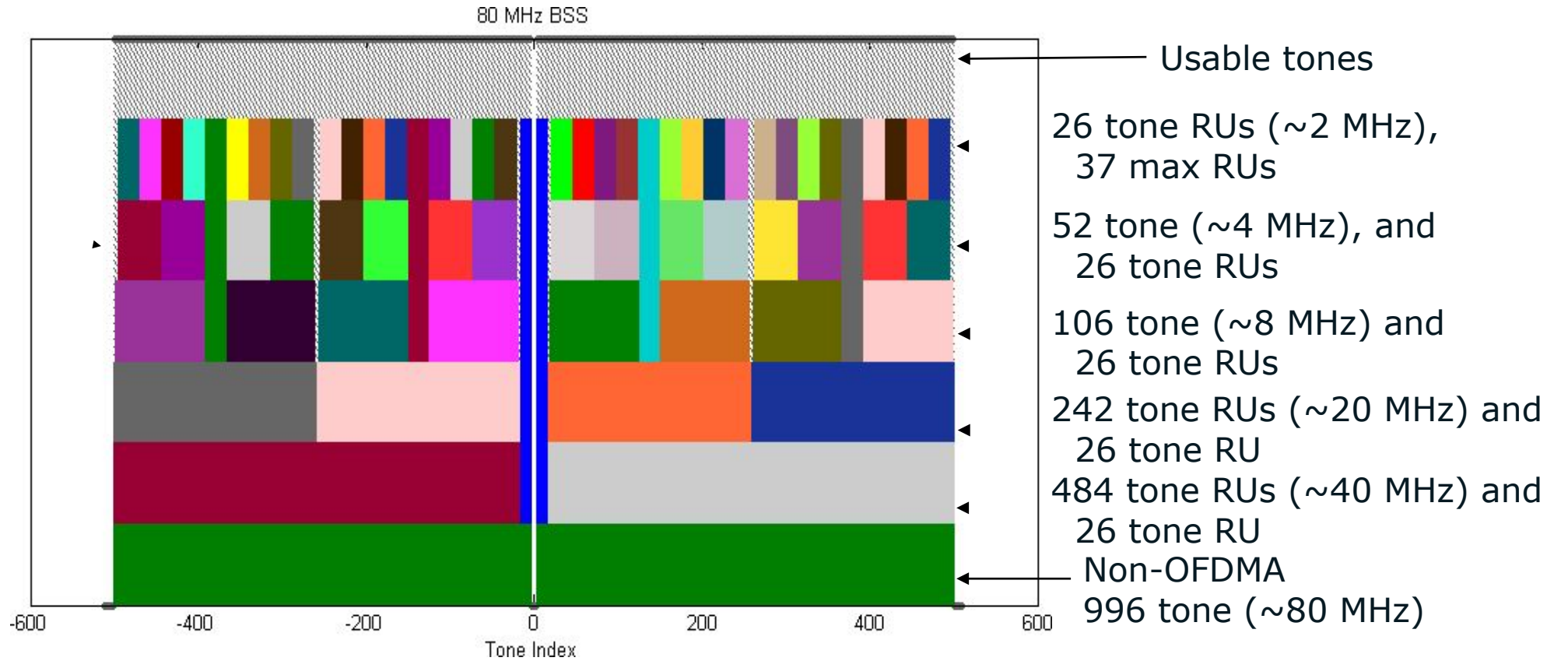
20MHz Resource Unit Split



THE MAXIMUM NUMBER OF RUS FOR EACH BANDWIDTH

RU type	20MHz	40MHz	80 MHz	160 (80+80) MHz
26-tone	9	18	37	74
52-tone	4 (+1)	8 (+2)	16 (+5)	32 (+10)
106-tone	2 (+1)	4 (+2)	8 (+5)	16 (+10)
242-tone	1	2	4 (+1)	8 (+2)
484-tone	NA	1	2 (+1)	4 (+2)
996-tone	NA	NA	1	2

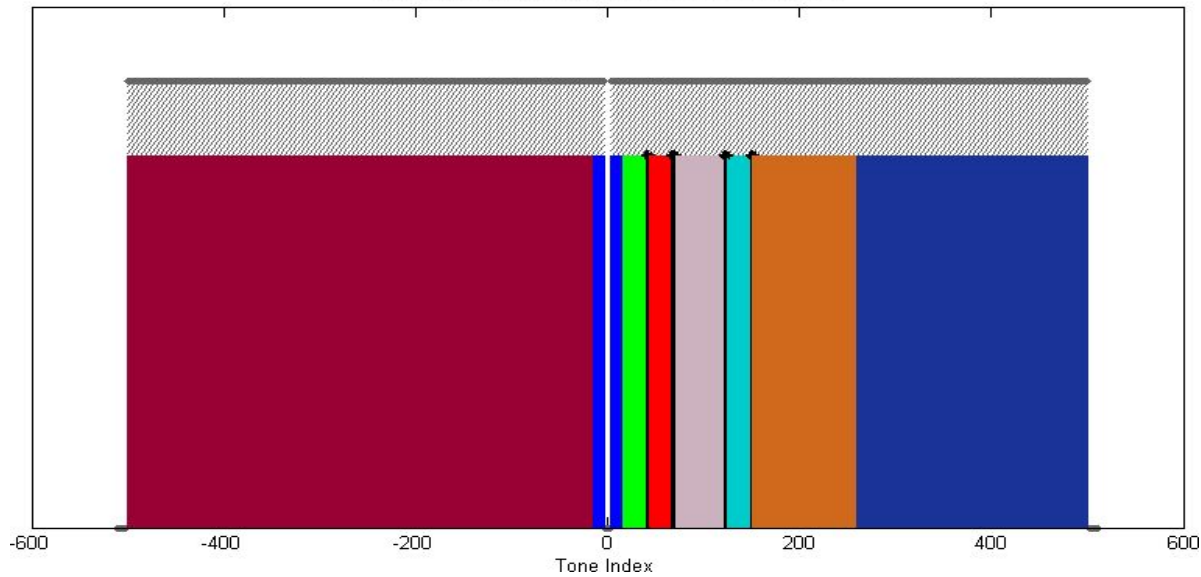
80 MHz BSS



OFDMA Resource Unit Allocation Examples

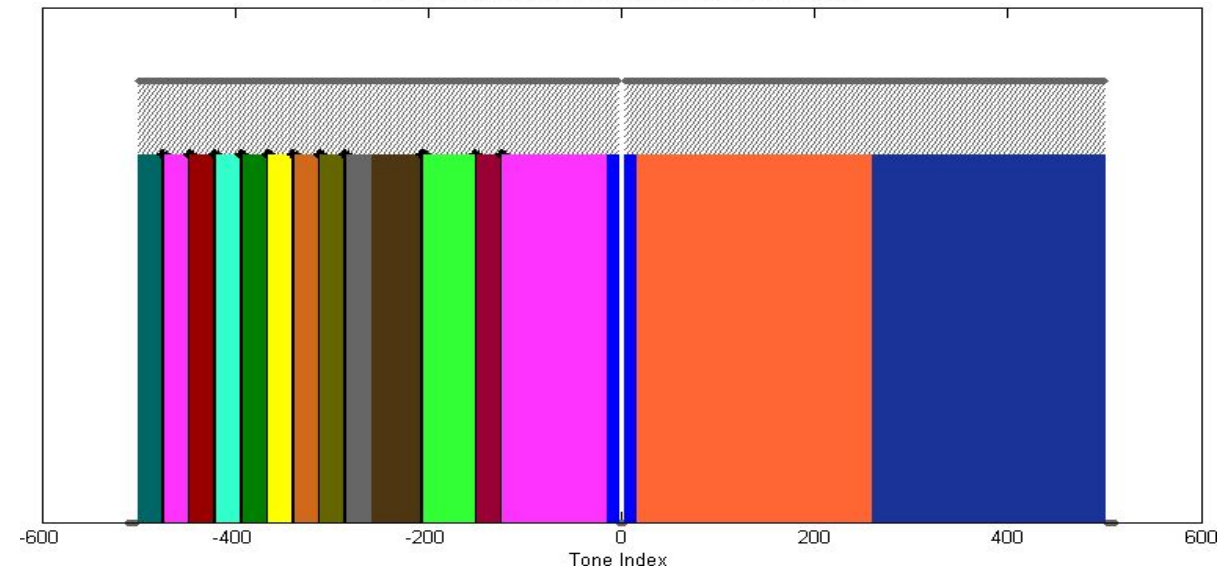
8 OFDMA assignments in 80MHz BSS

Example 2: 8 OFDMA assignments in 80 MHz BSS



16 OFDMA assignments in 80MHz BSS

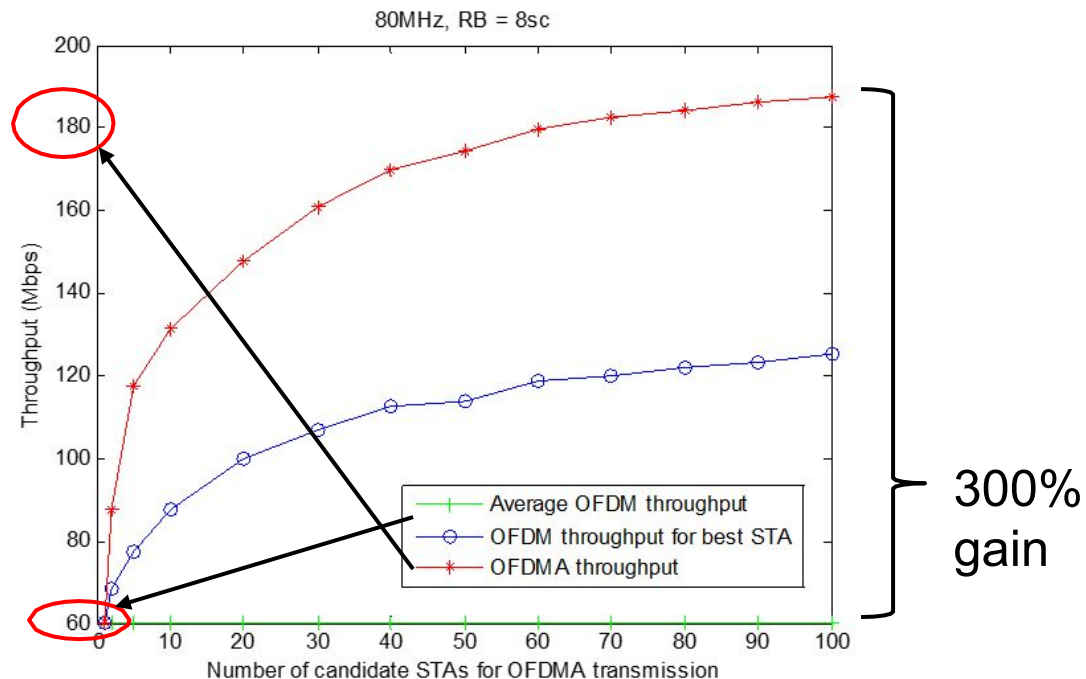
Example 1: 16 OFDMA assignments in 80 MHz BSS



RU assignments can vary packet to packet

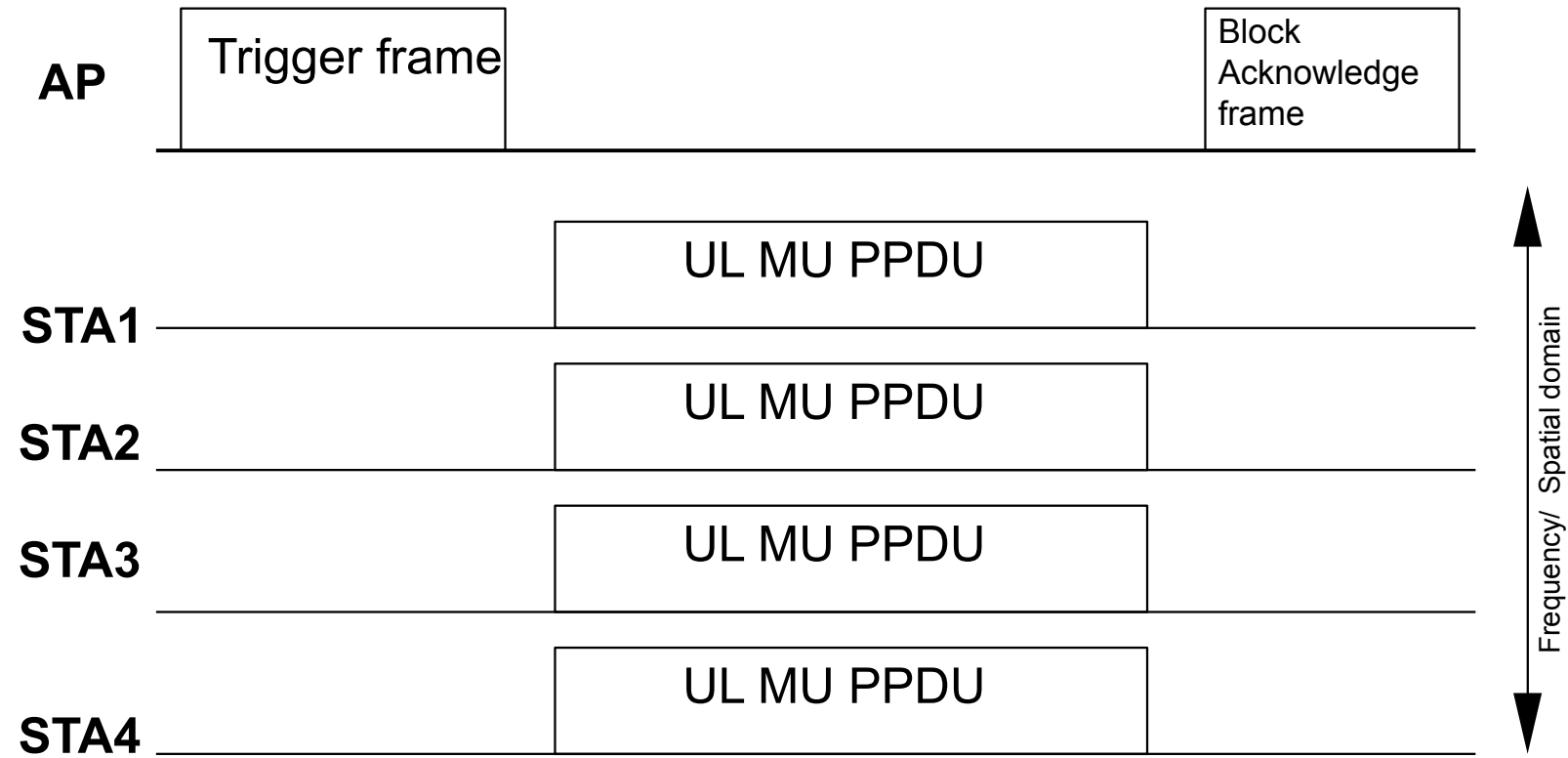
OFDMA throughput vs Number of STAs

- More active STAs achieve higher throughput gain
- The number of active STAs restricted by delay and overhead
- 10-20 active STAs can achieve quite good throughput gain
- We use 10 active STAs in the following simulations.



UL MU Operation

Basic Frame Exchange Sequence for UL MU transmissions



Basic Frame Exchange Sequence for UL MU transmissions

- New Trigger control frame
 - Specifies the length of the UL window
 - Specifies the users that may send during the UL window
 - Allocates resources for the UL-MU PPDU:
 - RU allocation
 - Spatial stream allocation
 - MCS to be used by the user
- UL MU transmission may be OFDMA or MU-MIMO
- Acknowledgement frame can be
 - DL MU transmission with individually addressed BlockAck frames
 - New “Multi-STA BlockAck” frame contained in Legacy frame or HE MU PPDU
- Trigger frame can be used as a Beamforming Report Poll, MU-BAR, MU-RTS, Buffer Status Report Poll, Bandwidth Query Report Poll...

